

roomplan: technical report (Applied AI Engineer case study, Round 2)

Repo: <https://github.com/KantiPT400/roomplan>. Every number below is regenerated by `scripts/reproduce.sh` and stored under `docs/benchmark/`. Max 6 pages (`docs/report.pdf`: 5 pages A4).

0. What was available, and what that means for every number

We were given three Stray Scanner LiDAR captures (no photos, no tape measurements): `single_room` (37 s), `single_scan_floor_only` (115 s) and `single_scan_with_ceiling` (215 s, 100 m walk). The last two are the **same apartment captured twice** (plan registration: normalised correlation 0.55-0.61 at 90 deg vs ≤ 0.28 at other rotations). We had no access to the rooms, so we could not tape-measure, stage damage, shoot photo folders or run a consumer scanning app. We therefore report three kinds of evidence and label each:

evidence	what it measures	where
synthetic ground truth	algorithmic accuracy and interval calibration against exact truth, on rendered captures with iPhone-like noise and drift	<code>scripts/synth_benchmark.py</code>
repeatability	two real captures of the same apartment; needs no truth	<code>scripts/benchmark.py</code>
reference	photo/video tier vs the LiDAR tier of the same capture (LiDAR itself unverified against tape)	<code>scripts/benchmark.py</code>

Nothing in this report is accuracy against a tape measure. That number is missing; we say so instead of inventing one.

1. Architecture

One command per capture (`python -m roomplan <folder>`); the tier is detected from the folder. All three tiers end in the **same geometry stage and output contract**: the video and photo tiers first turn their images into a “pseudo-LiDAR” capture (metric depth + poses in the Stray Scanner layout), so walls, rooms, openings, drift handling, damage and intervals are one code path.

```
LiDAR (Stray Scanner) —————
Video → frames → mono depth (Depth Anything V2-S, ONNX, CPU)
      → single-view metric/gravity per frame → pairwise
      registration (essential matrix, floor masked)
      → global heading/position solve → pseudo-LiDAR
Photos → same, all pairs instead of neighbours
```

→ drift pose graph → fuse depth
→ floor/ceiling → wall planes
→ free-space carving → rooms, openings, adjacency → polygons
→ intervals → damage → scope → JSON/PNG

Walls are vertical surfaces found from a 2D map of vertical extent: a 4 cm cell is wall if its points fill $\geq 60\%$ of 10 cm height bins up to the ceiling, or show ≥ 1.1 m of continuous vertical surface (walls seen only partly). Tables, sofas and counters fail both tests; full-height wardrobes and fridges pass (a known failure, section 7). Each wall plane is refined at 1 cm precision from its raw points; its standard error uses one effective sample per 4 cm of wall (points are spatially correlated). **Rooms** are free space carved by depth rays, cut by walls and by doorway-sized gaps between walls (0.55-1.40 m, jambs ≥ 0.5 m long); a gap is an opening only if free space is observed through it and different space lies on each side. Each opening is thus also an adjacency edge. An opening into observed space that is not a room must have ≥ 0.7 m of free space beyond it; shallower “openings” were niches, alcoves and a window bay (section 5). Door jambs are re-measured from raw points at handle height (cell-based ends were biased by up to 10 cm). **Room polygons** are rectilinear, and each edge snaps to a measured wall plane only if that plane was seen from inside the room (otherwise it is the neighbour’s face, one wall thickness away). Edges with no supporting plane (furniture-bounded, unseen) are drawn grey and carry ± 10 cm extra sigma. **Ceilings** are per room; a plane must cover $\geq 15\%$ of the room footprint (shelf and loft tops do not), and a room whose ceiling was never seen reports `not_observed` with a prior range, never a guess.



thick wall: plane measured, thin: inferred; dark: both ends measured, grey: an end bounded by furniture or unseen (wider interval)
red: opening width (m); lengths: value + 95% half-interval (m); narrow rooms: short walls unlabelled; all numbers in plan.json

Output on the supplied `single_scan_with_ceiling` capture (LiDAR tier): rooms with floor area and measured ceiling height, openings (red) with widths, every wall length with its 95% half-interval.

2. Tiers and device matrix

Full matrix: `docs/device_matrix.md`. LiDAR: iPhone 12 Pro and later Pro models, iPad Pro (Stray Scanner). Video and photo: any iPhone 15 or newer (native Camera). Capture protocol (one page, non-engineer): `docs/capture_protocol.md`; every instruction in it answers a failure seen in this data (return to start for loop closure; tilt to the ceiling once per room; floor in every photo; ~30 deg photo steps; doorway shots).

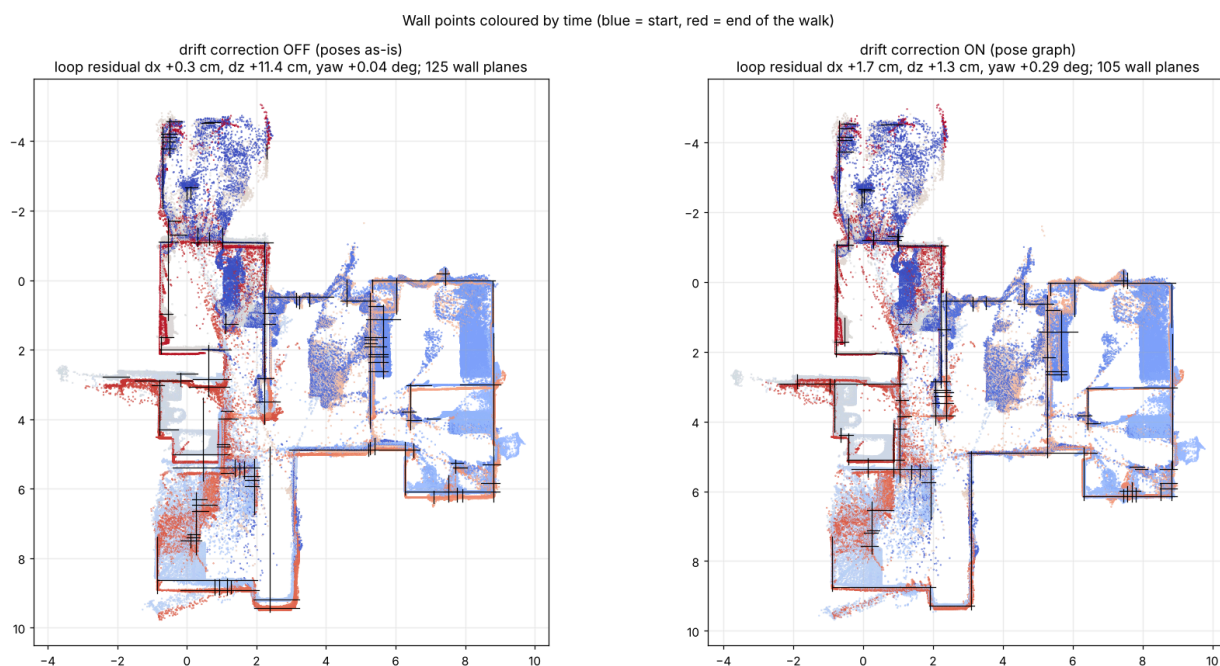
Video/photo metric scale comes from the camera height: the floor plane is found in each image's mono depth (lowest large near-horizontal plane; planarity fixes the disparity shift), and the camera is assumed to be 1.40 +/- 0.07 m above it (calibrated on the sample captures, whose LiDAR gives 1.41 and 1.46 m; sigma covers other operators). That 5% (1 sigma) is propagated into every video/photo interval. Registration uses two-view essential matrices with the floor masked out (repetitive tiles produced most false matches; section 6), translation length from metric depth, then a global solve: headings averaged over all edges, positions by robust least squares with all cameras at equal height.

3. Drift accountability

ARKit drifts in position over a long walk and occasionally relocalises with a jump (floor_only has two 17 m/s pose jumps). We cut the walk into 6 s segments (always at jumps) and solve a **pose graph whose edges are wall faces seen by two segments** (same axis, same observed side, overlapping >= 0.3 m). Pairs far apart in time are the loop closures. Huber-IRLS least squares, weak temporal tie (dropped across jumps), zero-mean gauge; floor height per segment as a unary term. Corrections are interpolated in time and applied to every pose. Heading is **not** corrected. A first version corrected each segment's yaw from its wall squareness, but real walls are not perfectly square, so that signal is biased, and our loop check cannot

resolve heading drift: the start/end heading difference moves by up to 1 deg with the window length alone (5/10/15% of the walk: -0.93, +0.04, -0.50 deg on the raw poses), more than the 0.45 deg the yaw version produced. With no trustworthy heading signal, we leave heading alone and report it. Ablation (--drift off; scripts/drift_ablation.py, figure docs/benchmark/drift_ablation.png), with_ceiling:

	poses as-is	pose graph
start/end loop residual (same wall faces)	dx 0.3 cm, dz 11.4 cm	dx 1.7 cm, dz 1.3 cm
start/end heading (10% windows; the estimate itself varies +/-0.5 deg)	0.04 deg	0.29 deg
wall-face spread (length-weighted median)	3.0 cm	2.5 cm
wall planes (duplicates from drift)	125	105



The drift graph's own edge residual (1.4 / 2.8 cm RMS) feeds a per-metre drift term into every interval.

4. Error budget and intervals

Every measurement reports a 95% interval, $\sigma^2 = \text{random}^2 + \text{systematic}^2 + \text{drift}^2$: * random: plane-fit standard error of each wall face (typically 2-8 mm), combined for both ends of a length; * systematic: LiDAR 5 mm + 0.4% of the length; video 1 cm + 1.5% and photo 1.5 cm + 2.5% (mono-depth shape), both plus the 5% camera-height scale term; * drift: per-metre term from the pose graph residual; * unsupported edge (end bounded by furniture or unseen): +10 cm; * ceiling: both planes' standard errors + systematic + a coverage term when the plane covers < 40% of the room; * damage extent: half a 2 cm cell along the region boundary + scale term.

5. Benchmark

Synthetic ground truth (6 rendered captures of a known two-room flat; smooth per-frame depth bias + mm jitter, 0.5% dropouts, slow pose drift, 17 deg rotation; docs/benchmark/synth_benchmark.json):

gate	result	intervals covering truth
rooms and adjacency	6/6 correct (2 rooms, 1 door)	
wall lengths (24)	24/24 within 2 cm , 17/24 within 1 cm, median 0.8 cm, max 1.8 cm	100%
wall-to-wall room dimensions (24; what a laser measure gives)	24/24 within 2 cm, median 0.8 cm	100%
ceiling height (12 rooms)	12/12 within 2 mm (gate 1.5 cm)	100%
door width (6)	6/6 detected, 5/6 within 2 cm, errors 0.8-2.2 cm, all undersized (bias)	100%

Repeatability, real data (with_ceiling vs floor_only, LiDAR, same apartment; benchmark_final.json). **The 1 cm / 0.5% gate fails.** Of 30 wall planes matched between the captures, 8 agree within 1 cm and 10 within 2 cm (median 2.7 cm); 8 of 45 plane-to-plane spans pass (median difference 3.8 cm); 1 of 8 matched room walls passes. The fitted relative scale between the captures is 1.002 (x) and 0.995 (z), up to 0.5%, which the LiDAR systematic term (0.4%) roughly covers; beyond that the differences are scattered, and the large ones are identity errors: a room edge on a full-height furniture front in one capture and on the wall in the other, and rooms that merge in one capture and split in the other. Our intervals cover the cross-capture wall difference 25% of the time: they are calibrated for measurement error (synthetic: 100%) but not for “which surface is the wall”, the real-world error mode.

Door widths, real data (same two captures; there is no tape truth, so this is agreement, not accuracy): 13 and 15 openings; 7 matched across captures; **2/7 agree within 2 cm** (4/7 within 2.2 cm, median 2.2 cm, intervals cover 57%). Inspecting the raw points of each disagreeing pair: a door left ajar (0.50 vs 0.67 m), a cabinet against a jamb in one capture (0.71 vs 1.07 m). Before the change below, two more “doors” were a niche 0.3 m deep and an alcove 0.5 m deep, disagreeing by 4-9 cm; a one-room opening must now lead into ≥ 0.7 m of space (removed 0.18-0.63 m deep recesses, kept 0.99-1.47 m). Measuring jambs higher up to avoid furniture was tried and was worse (7-13 cm). The protocol now says to open every door fully.

Ceiling: only with_ceiling looked up; 11 of its 12 rooms have measured ceilings (1.97-3.08 m; the 1.97 m one is flagged “below 2.1 m: may be a loft or storage platform, verify”; intervals ± 2 -3 cm). The cross-capture ceiling spread cannot be computed: floor_only never saw a ceiling and correctly reports not_observed rather than guessing.

Staged damage (scripts/inject_damage.py): a soft yellow-brown stain (0.114 m^2) and a 0.60 m hairline crack painted onto a measured wall in every frame using the capture’s own depth and poses. Detected on the right wall with the right classes: crack 0.62 m [0.54, 0.70] (truth 0.60); stain 0.141 m^2 [0.111, 0.172] (truth 0.114, inside; the detector overestimates); scope lines “rake out, fill and repaint crack 0.62 m” and “stain-block primer + repaint 0.95 m^2 ” on that surface. On the unmodified single_room and floor_only captures it reports 0 regions and 0 flags (it reported 26 regions before its precision gates). with_ceiling’s video never reached our machine, so damage was not run on it with real colour; a stand-in video of the same length ran the stage end to end without error (108 frames, 42 walls; docs/sample_outputs/README.md).

Photo and video tiers vs LiDAR (same capture; benchmark_final.json, plans in docs/sample_outputs/):

	photo (55 photos, 7 room folders, floor_only)	video (single_room clip, 37 s)
footprint vs LiDAR	0.958 (gate $\pm 8\%$: passes)	0.58
rooms (LiDAR has) / matched	7 (11) / 2	4 (3) / 1
images in the mapped component	43 / 55	43 / 223 frames
wall intervals	± 0.20 -0.30 m (scale prior + unsupported ends)	± 0.21 -0.30 m

Neither passes its full gate: room structure and adjacency are not yet right. The photo set had to be cut from a LiDAR walkthrough (no photo folders were supplied) and the clip turns faster than our video protocol allows (p99 137 deg/s), so registration breaks into separate stretches; only the largest is mapped, and the output says how much of the input it used. The footprint intervals are wide on purpose: “confident garbage” is the failure we designed against.

Reproduction and stability. scripts/reproduce.sh on a fresh clone of the final commit with requirements-lock.txt (42 min on the 2-core VM) regenerated every number in this report bit for bit, photo and video tiers included (docs/benchmark/benchmark_fresh_clone_locked.json). An earlier fresh install without the lock file (benchmark_fresh_clone_ort1.30.json) also matched every LiDAR, drift, synthetic and staged-damage number, but it had pulled onnxruntime 1.30 instead of 1.29; its mono depth differs by $\sim 2e-7$ (relative), and the photo tier then merged three rooms into one (7 $\rightarrow$ 5 rooms, footprint 0.958 $\rightarrow$ 0.937) and the video tier gave 3 rooms (footprint 0.74). The registrations had the same structure (109 pairs, components 43 + 2); the two photo plans differ mainly by one wall segment, present in one and missing in the other, which merged three rooms. So the photo/video room partition is not stable under tiny perturbations, which is the same weakness as their failing room counts. requirements-lock.txt pins the versions behind the reported numbers.

Robustness (a crash would score zero on a walk-in capture): the first 20% of a walk and a 10 s LiDAR scan give partial plans; 2 photos of one room, 3 folders of 2 non-overlapping photos and a 5 s clip give single_view_estimate rooms (one photo’s extent, 30% sigma, no adjacency) instead of nothing; a parent

folder of a Stray Scanner export is resolved to the capture inside it; a failure in damage detection is recorded in meta and the plan is still written.

Head-to-head vs a consumer app: not done. It needs a LiDAR iPhone in the benchmark rooms; we had neither. **Timing** (2-core VM, CPU): LiDAR 16 s (single room) to 60-75 s (115-215 s walks; damage frames included); photo tier 7-11 min for 55 photos; video tier 14-18 min for a 37 s clip at 6 fps (mono depth dominates). Clean machine to first LiDAR result: 93 s.

6. The fix loop (full story in docs/fix_loop_declaration.md and fix_loop_result.md)

Worst gate: photo-tier stitch (footprint 0.54x LiDAR, 0/8 rooms). Hypothesis, with evidence from scoring every photo registration against the capture's own poses: false registrations caused by repetitive floor tiles (157 of 203 off by > 15 deg; their inliers 71% on the floor vs 7%; they join photos 4.5 m apart). We committed the declaration before the fix, then found and fixed a bug in our own truth checker and corrected the declaration's numbers (before running the fix). Fix: mask floor features, reject registrations whose transported depth disagrees with the target's own depth. **Result:** global solve went from self-contradictory to consistent (heading residual 36.5 $\rightarrow$ 3.3 deg, position 0.23 $\rightarrow$ 0.09 m), false registrations 157 $\rightarrow$ 67, but rooms matched 0 $\rightarrow$ 1 and footprint 0.54 $\rightarrow$ 0.52: **the prediction was badly wrong**. Post-mortem: a second false-match source (one photo with a repetitive pattern acting as a hub) and, more importantly, a downstream stage we had not checked: mono-depth walls rarely passed the wall test. The partial-wall rule added afterwards (section 1) came from the synthetic benchmark; hub suppression is in, guarded so it cannot split the graph. Acting on the post-mortem (coarser wall cells for mono depth, rooms grown into space seen once; commit e283758) brought the photo footprint to 0.958 and rooms to 7 of 11; the room structure still fails.

7. Known failure modes (and what the output does about them)

- **Full-height furniture** (wardrobes, fridges) is a wall to geometry; room edges may land on its front. Seen in the repeatability data. Next: the top of the surface vs the ceiling, and colour/textural discontinuity.
- **Room partitioning between captures** is not yet stable (merges/splits around wide openings).
- **Doors left ajar** hide part of the doorway: one door measured 0.50 m in one capture and 0.67 m in the other (leaf ~ 15 deg off the wall plane). The protocol now says to open every door fully.
- **Mirrors and glass** (the sample bathroom has a glass shower screen): LiDAR returns the reflection or the surface behind; mirrored geometry appears beyond the wall and is cut by the wall/free-space test only when the real wall is also seen. Protocol: close mirrored wardrobes, do not stand still facing a mirror.
- **Wet-look / glossy floors:** tile reflections create false SIFT matches (the fix-loop failure) and mono depth under-estimates glossy floors; floor features are masked in registration.
- **Low light:** LiDAR depth is unaffected; video/photo tiers lose features and mono depth degrades. The protocol asks for all lights on; otherwise use the LiDAR tier.
- **Textureless white walls:** COLMAP fragmented into 7-17 models; replaced by pairwise registration, still the main video/photo weakness.
- **Ceilings never seen** are reported not_observed, never guessed. **Sloped ceilings, curved walls and non-Manhattan layouts** are outside the model (polygons are rectilinear).
- **Damage detector** is rule-based (no labelled data): it detects yellow-brown stains, dark speckled mould-like regions, thin cracks and off-plane surface loss on measured walls only; grey smoke stains, damage on floors and ceilings, and damage behind furniture are not detected.

8. Models, data and disclosure

Pretrained models: Depth Anything V2 Small (ONNX export, fetched and checksum-verified by scripts/fetch_models.py, or .sh); SIFT (OpenCV). pycolmap drove the first video-tier version and is no longer required (roomplan/sfm.py is kept for the record of why it was replaced). No network calls at run time. Raw benchmark data: the three sample captures (supplied); everything derived is regenerated by scripts; synthetic captures are rendered by scripts/synth_capture.py from fixed seeds.